

Power supply with the same power PIN, the higher switching frequency fSW is, the smaller IPK MOSFET is; in this way, the low current MOSFET can be used which will save more cost for user. To further verify above view, we test the VDS, IDS stress of two MOSFETs based on two different switching frequency.

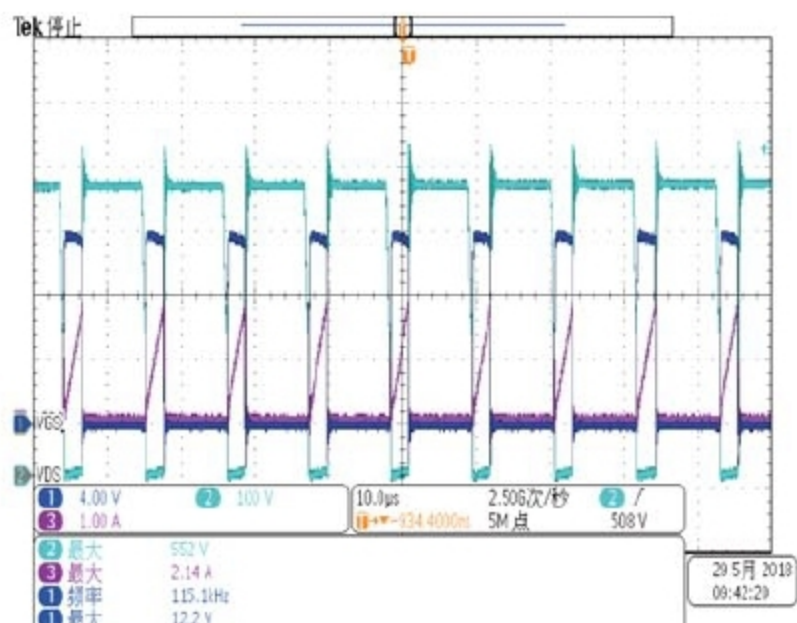


Fig.2 36W LED power Flyback Fsw=115KHz IPK MOSFET=2.14A

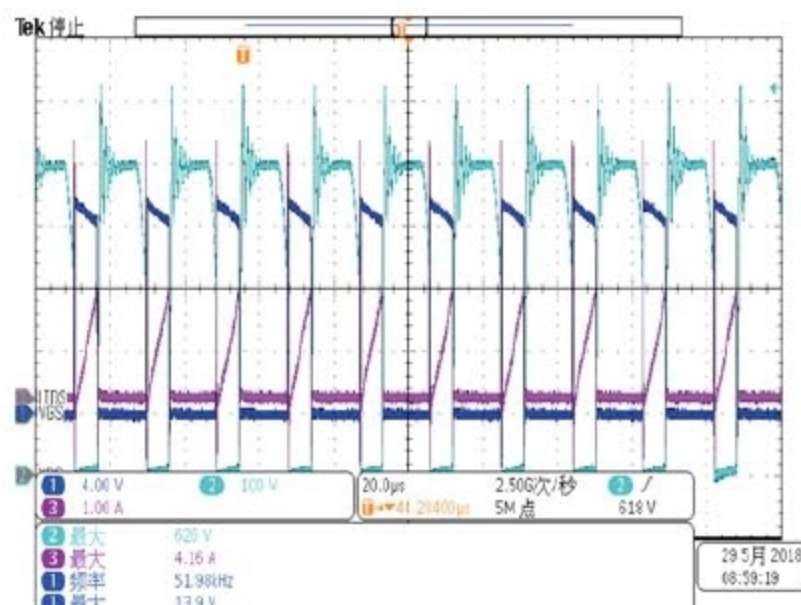


Fig.3 36W LED power Flyback Fsw=51KHz IPK MOSFET=4.16A

b. The high frequency transformer can be smaller size and lower cost

The primary side inductance Lp of transformer with IC controller UC3842 CCM is calculated as below:

$$L_p = \frac{1}{2} \times \frac{(V_{BULK(min)})^2 \times \left(\frac{N_{PS} \times V_{OUT}}{V_{BULK(min)} + (N_{PS} \times V_{OUT})} \right)^2}{0.1 \times P_{IN} \times f_{SW}} \quad \text{[formula 6]}$$

From formula 6, fSW is negative proportion to Lp; the lower Lp is, the less transformer winding turns NP is, and the magnetic core size can be smaller and thinner. The ultra smaller size Power Supply become possible to be designed. The more important point is that the cost of more compact transformer is lower.



Fig.4 36W LED power Flyback Fsw=115KHz
PQ transformer size: 26.5mm*26mm*19

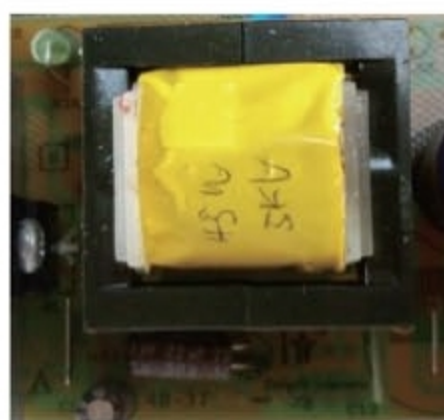


Fig.5 36W LED power Flyback Fsw=51KHz
EE transformer size: 32.5mm*30mm*20mm

Summary

Comparing with VD-MOSFET, SJ-MOSFET has smaller Ciss or Qg, smaller turn-off loss, which is more suitable for high switching frequency. If MOSFET at high frequency power supply primary side, there will be a smaller current stress, lower current specification components can become optional; In the mean time, the transformer in high frequency power supply can be smaller size and lower cost.

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